IMS560 Advanced Database Management System

Group Project Assignment (60%)

Project Overview

In this project, each group will design and develop a fully functional software system — **either a website or a mobile application** — that integrates with a **SQL or NoSQL database**. The system must allow end-users to perform **CRUD (Create, Read, Update, Delete)** operations on the data stored in the database.

This assignment assesses students' ability to apply advanced database concepts to real-world system development, including database design, implementation, query optimization, API/database connectivity, and documentation.

Learning Objectives

By completing this project, students will be able to:

- Design relational or non-relational databases based on system requirements.
- Implement CRUD operations using appropriate data models and queries.
- Develop a functional application with front-end and back-end integration.
- Demonstrate best practices in database security, indexing, and performance.
- Prepare a professional technical report and deliver a system presentation.

Project Requirements

A. System Development

Each group must build a system with the following minimum requirements:

1. Application Type

- A **web-based system** (HTML/CSS/JS + PHP, JSP, ASP, etc.) **OR**
- A **mobile application** (Android Studio, FlutterFlow, Ionic, XCode, etc.)

2. Database

Choose **either:**

- **SQL database** (MySQL, PostgreSQL, SQL Server, MariaDB, SQLite, etc.)
- **NoSQL database** (MongoDB, Firebase, Cassandra, Redis, etc.)

3. Core Features

Your system must include:

- At least **two user roles** (e.g., admin + normal user)
- At least **three main modules/functions**
- Fully functional **CRUD operations** for all major data entities
- Secure user authentication (basic login or stronger)
- Proper validation and error-handling
- Clear UI for interacting with the data
- Implementation of indexing / query optimization where applicable

4. Additional Recommended Features (optional for bonus marks)

- Search or filtering functionality
- Data visualization (charts, reports)
- Cloud database deployment
- REST API design

Deliverables

A. System Presentation (10%)

A 10–15 minute group presentation covering:

- Introduction and motivation
- Database design and implementation
- Technology stack
- System demonstration
- Challenges and solutions

B. Final Report (15%)

Your written report must include:

1. Introduction & Planning

- Project background
- Objectives
- Scope
- Group roles & responsibilities
- Tools and technologies used
- Gantt chart or timeline

2. System Analysis

- Problem statement
- Intended users
- Functional & non-functional requirements
- Use case diagrams / user stories

3. System Design

- System architecture
- ERD (if SQL) or data model/design schema (if NoSQL)
- Data dictionary or collection structure
- Wireframes / UI design
- Sequence diagrams (optional)

4. Database Implementation

- SQL scripts / NoSQL collections
- Sample queries (CRUD + any advanced operations)
- Security measures
- Justification of design decisions

5. System Testing

- Test cases

6. Conclusion

- Limitations
- Future improvements

C. Working System / Source Code (35%)

You must submit:

- Complete source code
- Database export or JSON dump

Group Formation

- Each group should consist of **2–4 students**.
- Groups must inform their project title and chosen database type with the instructor.

Suggested Project Ideas

- Inventory management system
- Online booking/reservation system
- E-commerce mini-platform
- Student or staff management system
- Event management system
- Fitness or habit tracking app
- Library management system
- Food ordering application

(You may propose your own idea.)